

3 (a) (i) 0.8 cm B1 [1]

(ii) (max.) kinetic energy = 2.56 mJ C1

$v_{\text{(MAX)}} = \omega a$ C1

(max.) kinetic energy = $\frac{1}{2}m\omega^2 a^2$ or $\frac{1}{2}m\omega^2 (a^2 - x^2)$ C1

$2.56 \times 10^{-3} = \frac{1}{2} \times 0.130 \times \omega^2 \times (0.8 \times 10^{-2})^2$ M1

$\omega = 24.8 \text{ rad s}^{-1}$ C1

$f = \omega/2\pi$ M1

= 4.0 Hz (3.95 Hz) A0 [6]

(b) (i) line parallel to x-axis at 2.56 mJ B1 [1]

(ii) 1 4.0 Hz B1

2 0.50 cm (allow $\pm 0.03 \text{ cm}$) B1 [2]

4 (a) (i) either lines directed away from sphere